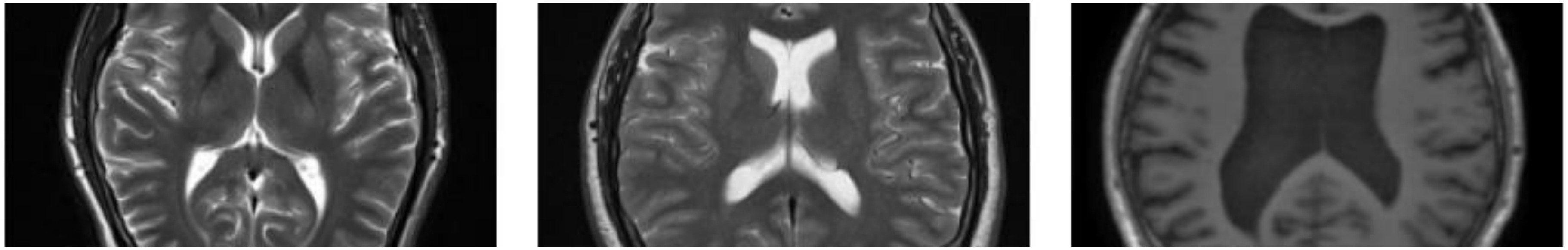
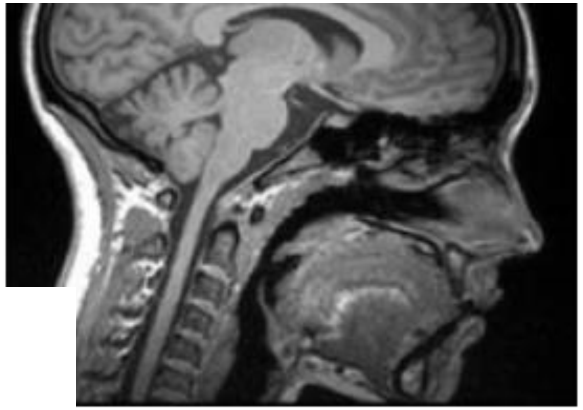
Universität Zürich[UZH]

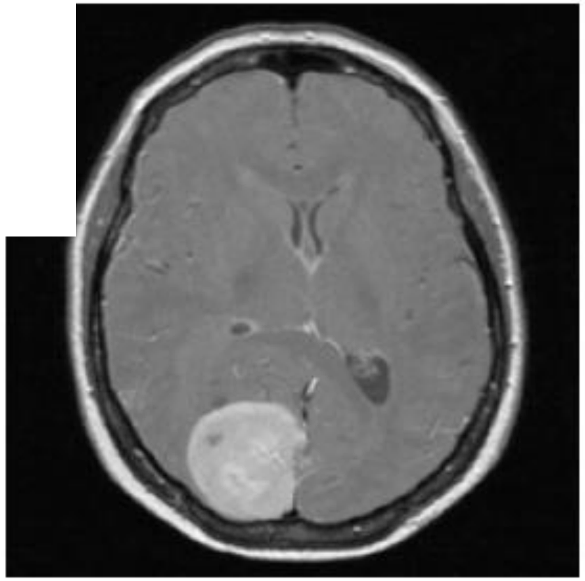
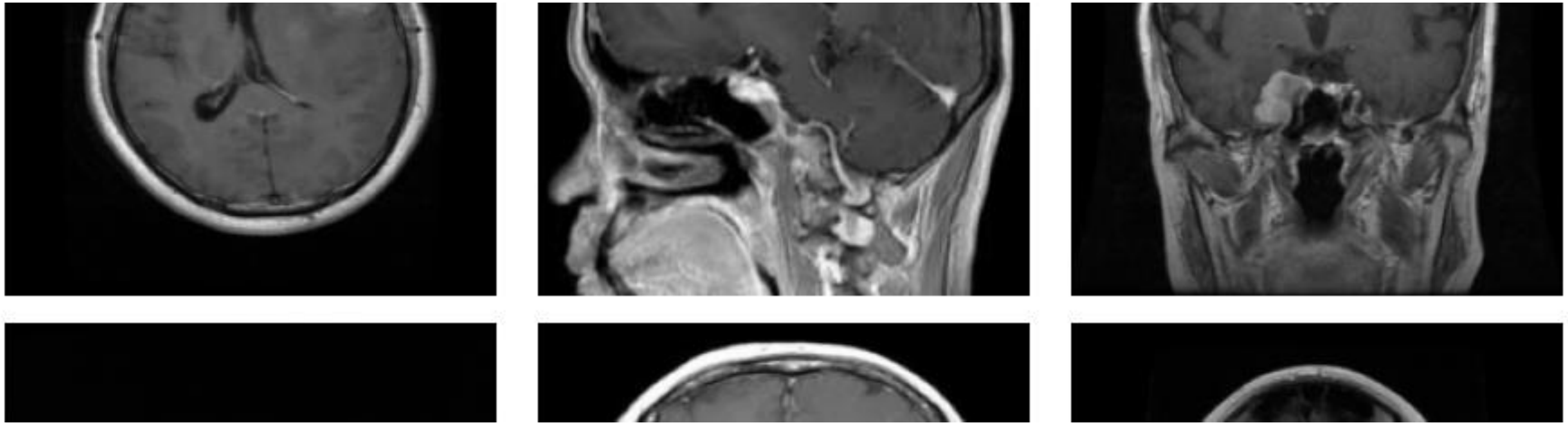
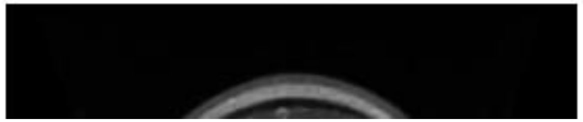



Brain Tumor Classification with Vision Transformers

Investigating the performance of polar coordinate positional embeddings (PoPE)

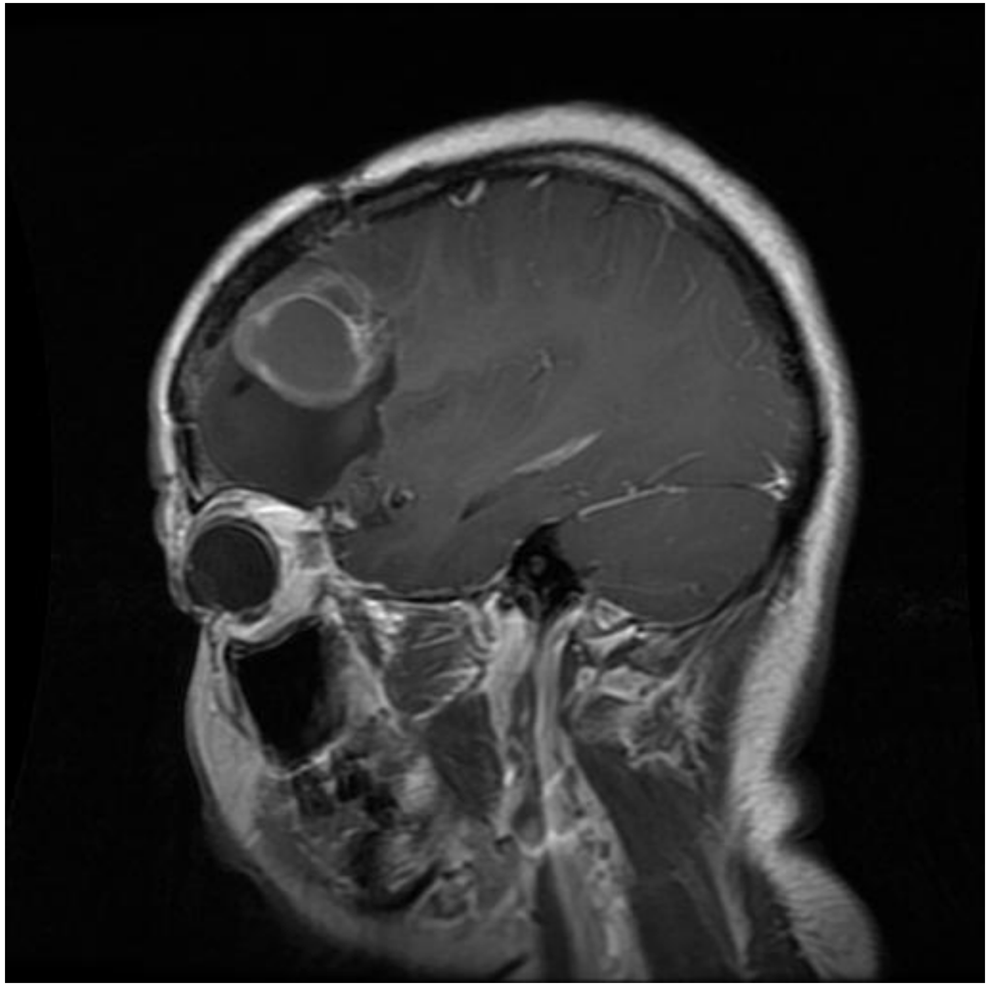
AML2026 Group 09:

Leonard Wagner, Valentina Jordan, Valentina Zingarello, Fabio Bucher

Brain Tumor Classification

- Most serious neurological disease
- Early diagnosis is critical
- Magnetic Resonance Imaging is the state of Art
- Manual MRI analysis is challenging



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Brain Tumor Classification and Deep learning

- CNNs used traditionally
- CNNs capture local image features
- Transformers might capture long-range spatial relationsships

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Brain Tumor Classification and Deep learning

- CNNs used traditionally
- CNNs capture local image features
- Transformers might capture long-range spatial relationsships

- Transformer-based glioma MRI analysis shows:
  - Large receptive field
  - Capture relationships between tumor and surrounding tissue
  - Global contextual information learning through self-attention mechanisms

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Our Project

Task Description:

- Multi-class brain tumor MRI classification
  - For a given MRI image, the goal is to predict a label $y \in$ {glioma, meningioma, pituitary, no tumor}

The Proposed Model:

- Vision transformer encoder with PoPe (Polar Coordinate Positional Embeddings)
  - PoPe might help disentangle
    - Positional information (where is the tumor)
    - Content information (what does the tumor look like)
- Encoder outputs are passed to an MLP classifier

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Our Project

Evaluation Protocol:

- Hold-out testing
- Two baselines:
  - Vision transformer with RoPE as positional encoding
    - ➢ Allows for a targeted evaluation of the effect of PoPE
  - ResNet18
- AUROC as main metric

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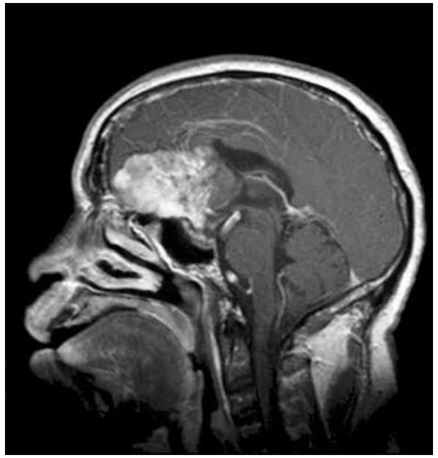
Our Project

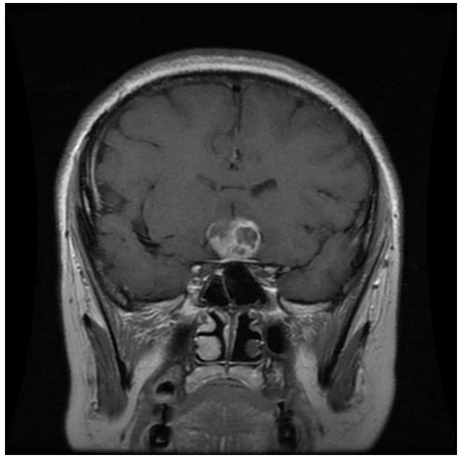
- Publicly available brain MRI dataset from Kaggle
    - contains multiple clinically relevant tumor categories
    - heterogenous scans in terms of shape, location and growth pattern of tumors

- 4 categories of scans:
    - Glioma (N = 926)
    - Meningioma (N = 937)
    - Pituary Tumor (N = 932)
    - No Tumor (N = 469)

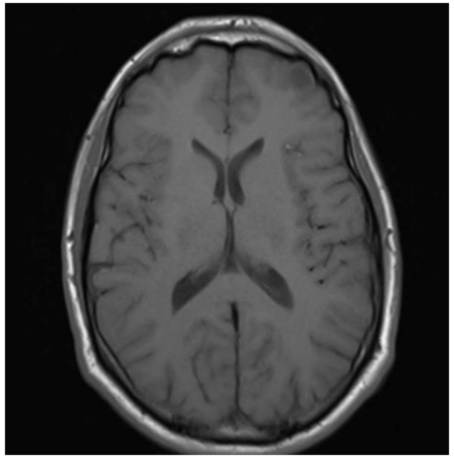


Glioma

Meningioma



Pituary Tumor



No Tumor

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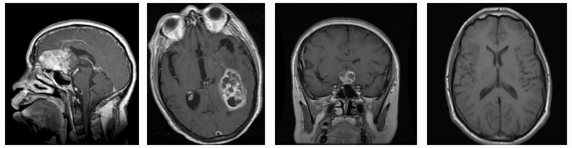
Data Pipeline & Augmentation

RAW POOL

Brain MRI — labelled images

Original Training + Testing folders merged, then re-split.



STRATIFIED SPLIT · PER CLASS SHUFFLE

Train · Val · Test

train 80%	20%	20%

Label distribution held identical across all three splits. Test split = 325 scans.

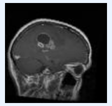
CLASS IMBALANCE HANDLING

Inverse-frequency class weights

Normalised so mean weight = 1, fed into weighted cross-entropy.

TRAIN TRANSFORMS · STOCHASTIC

VARIANTS_PER_IMAGE ×3



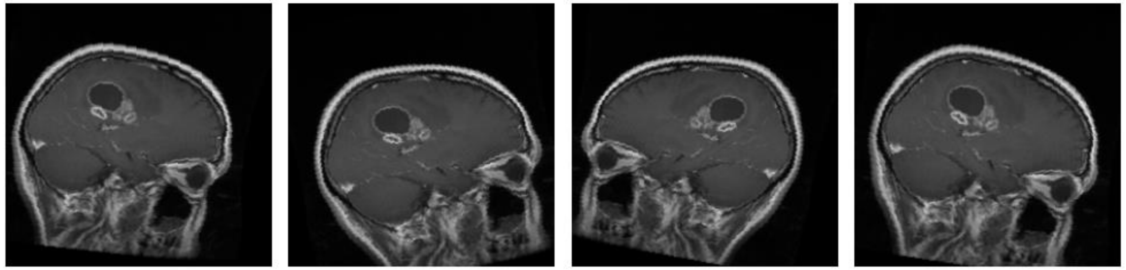
- **Resize 224 × 224**
  bilinear, antialias
- **EnsureRGB**
  1-ch → 3-ch · drop alpha
- **RandomAffine** AUG
  ±15° · translate 5% · scale .95–1.05 · shear 5°
- **RandomHorizontalFlip** AUG
  p = 0.5
- **ToDtype float32**
  scale to [0,1]
- **Normalize**
  per-split μ, σ

rotate

flip

shift

Augmented variants — glioma_tumor



VAL / TEST TRANSFORMS · DETERMINISTIC

- **Resize 224 × 224**
  bilinear, antialias
- **EnsureRGB**
  1-ch → 3-ch
- **ToDtype float32**
  scale to [0,1]
- **Normalize**
  per-split μ, σ

DATALOADERS

Batched tensors → models

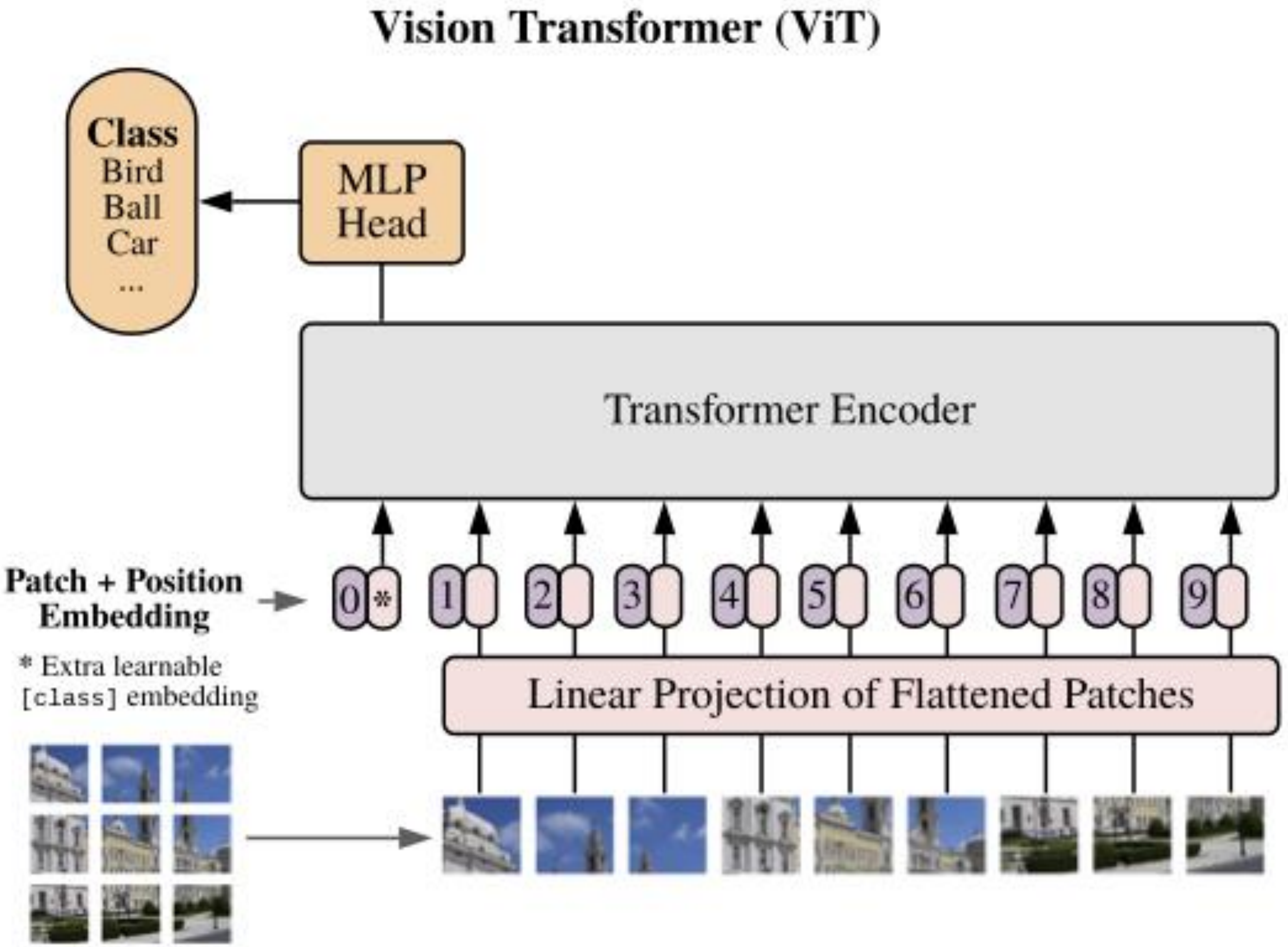
batch 32 | train: shuffle | val/test: ordered

Output: (B, 3, 224, 224) float32, normalized.

Vision Transformer

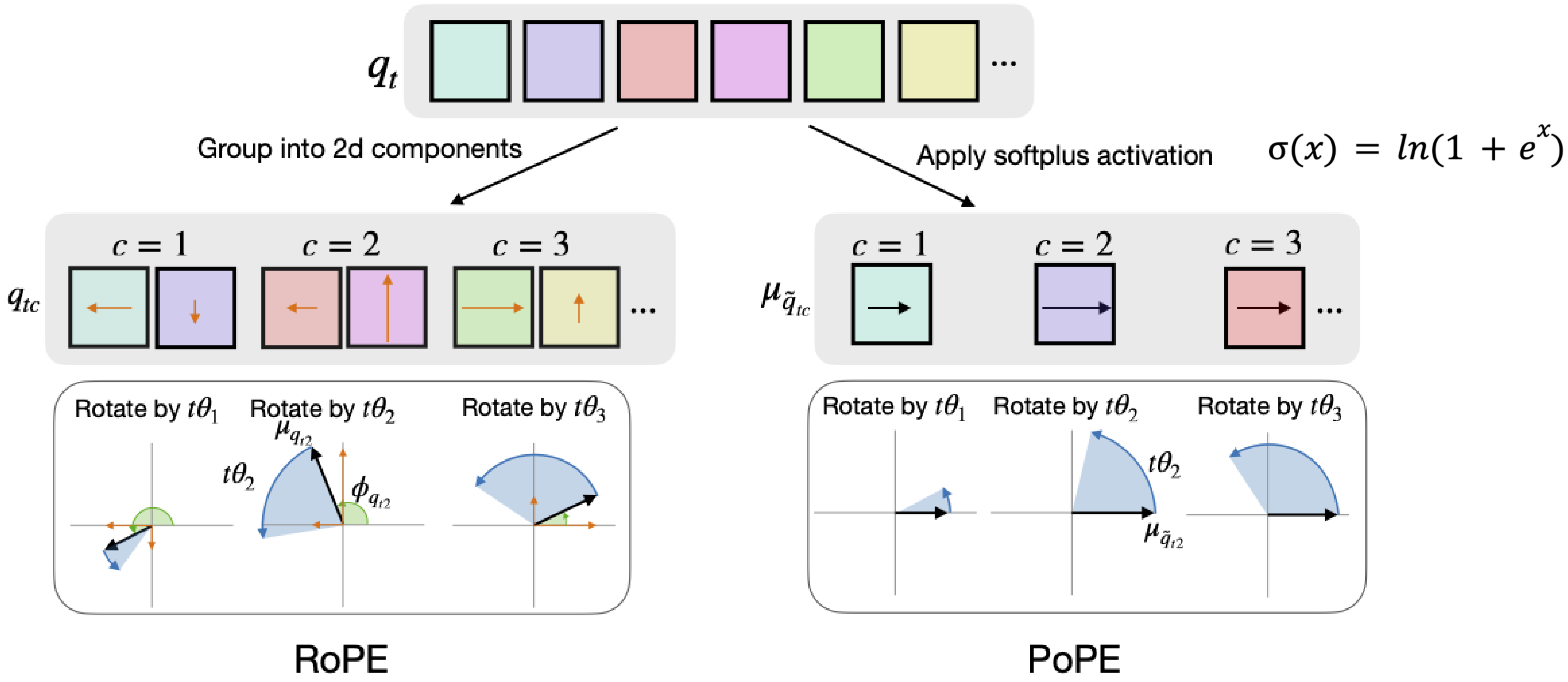
A Vision Transformer can be used for image classification by:

- reshaping the image into a sequence of flattened 2D Patches
- adding positional embeddings
- feeding the embedding of special token `[class]` to an MLP



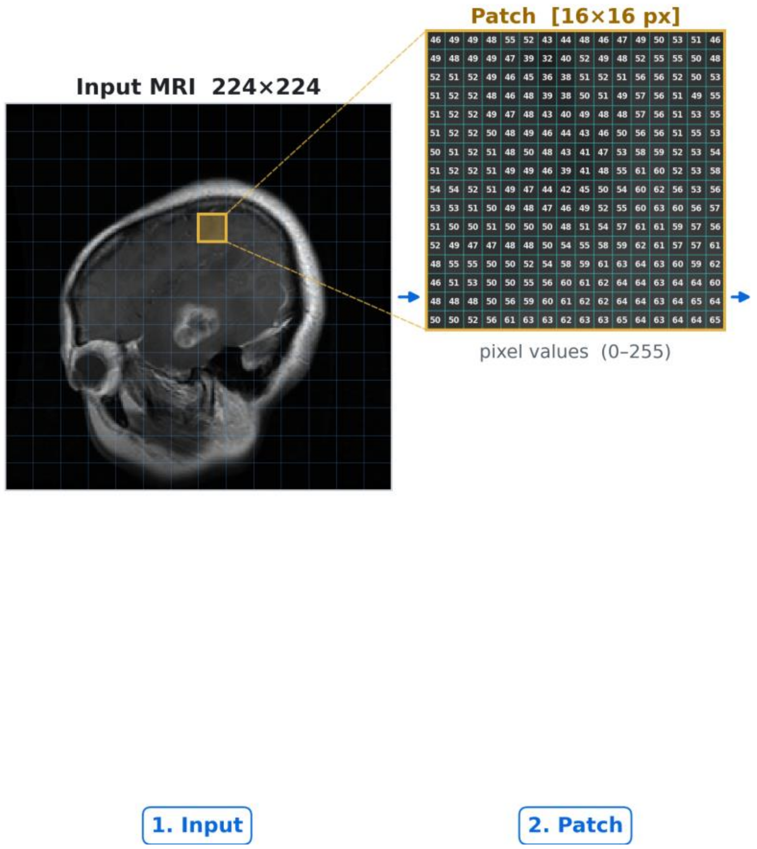
Source: Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv. https://doi.org/10.48550/ARXIV.2010.11929

The Intuition behind PoPE

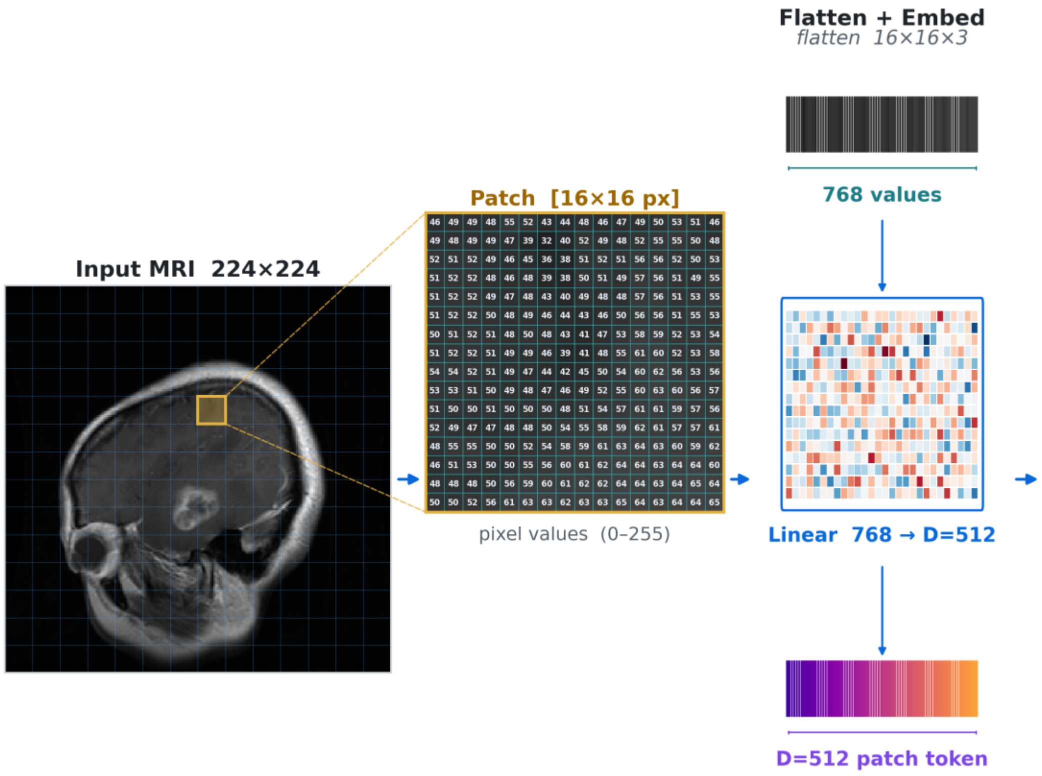


Source: Gopalakrishnan, A., Csordás, R., Schmidhuber, J., & Mozer, M. C. (2025). Decoupling the 'what' and 'where' with polar coordinate positional embeddings. arXiv (Preprint). https://doi.org/10.48550/ARXIV.2509.10534

Forward Pass of ViT – Giloma Tumor Example



Forward Pass of ViT – Giloma Tumor Example

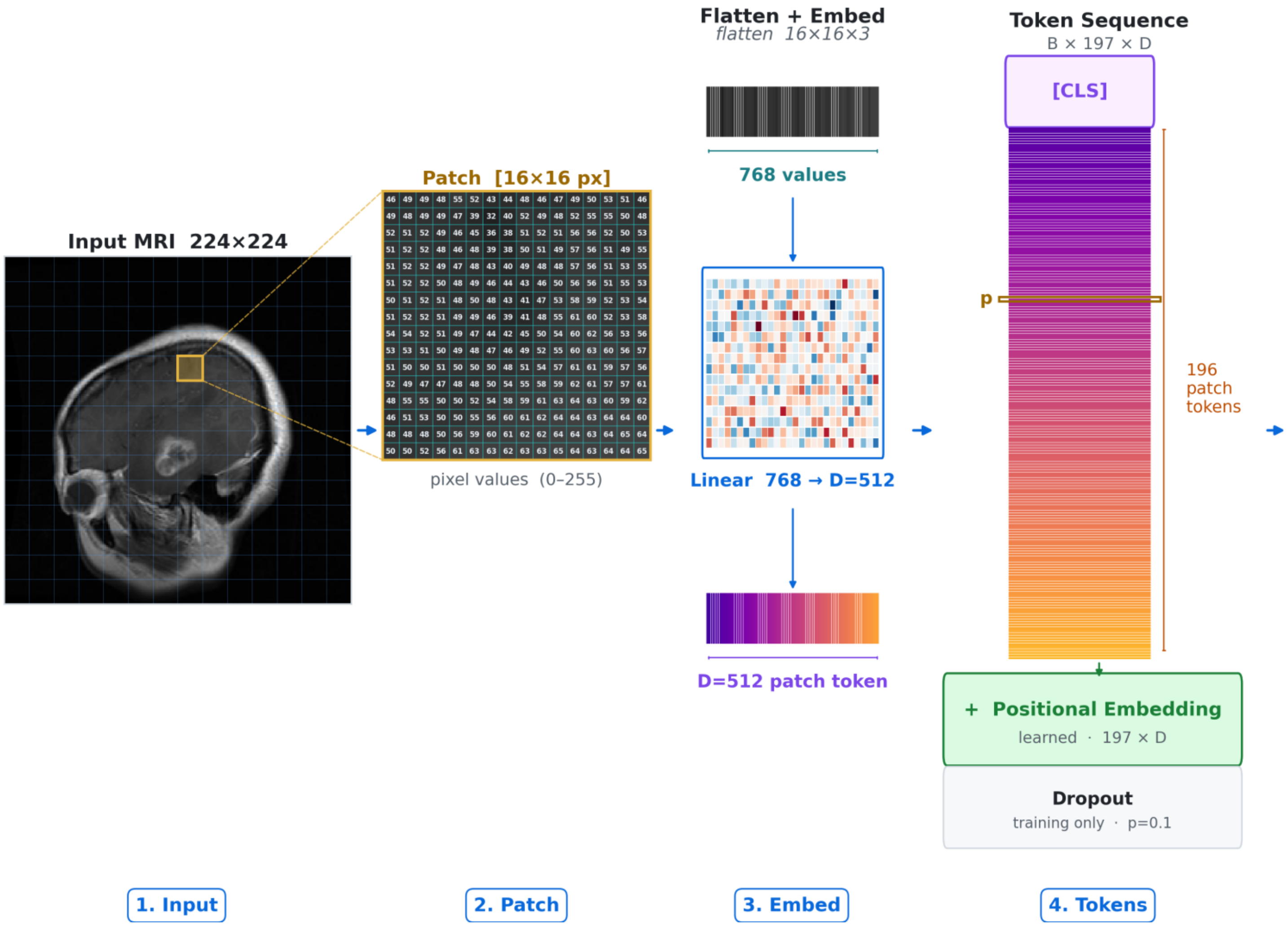


1. Input

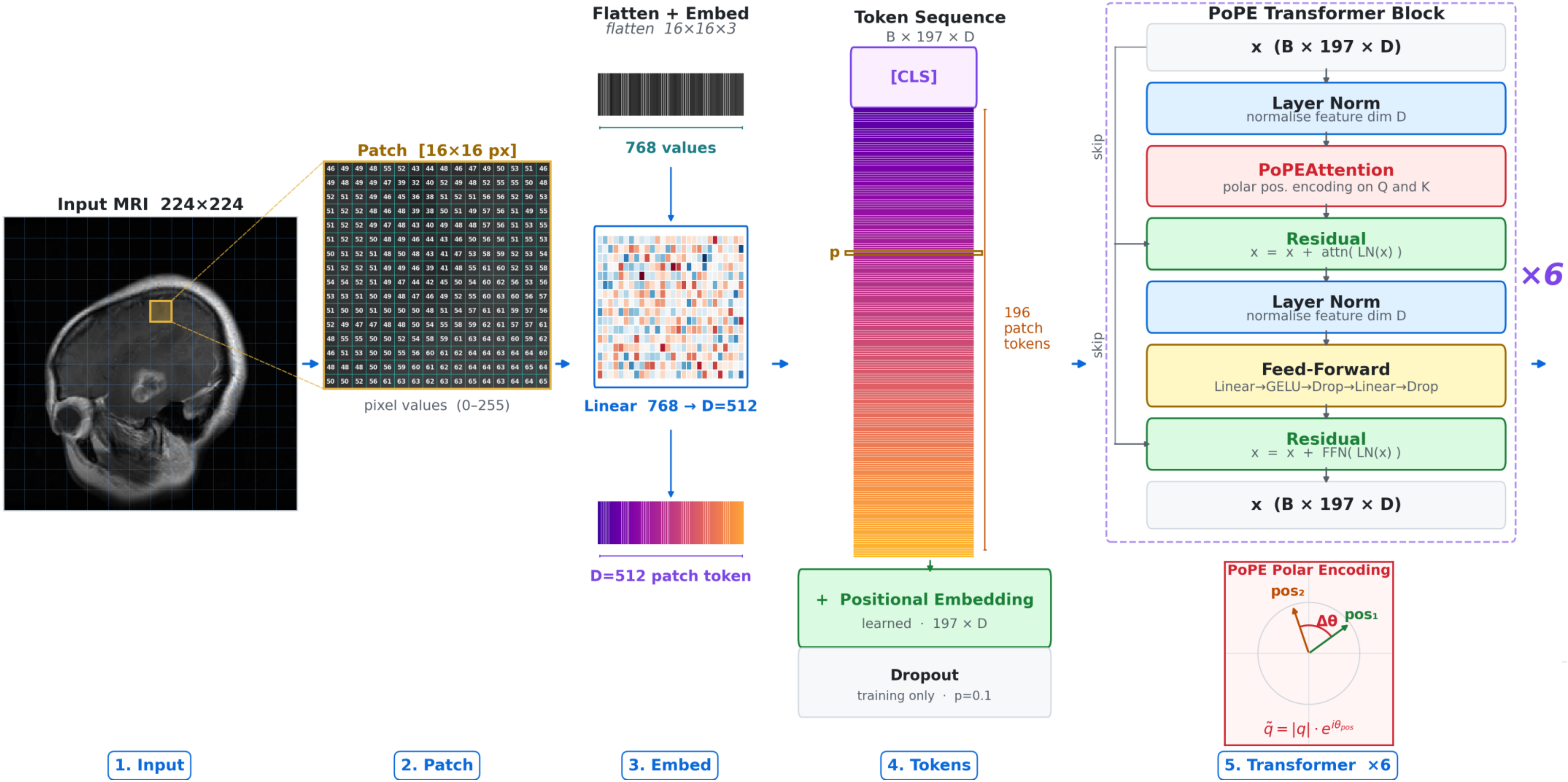
2. Patch

3. Embed

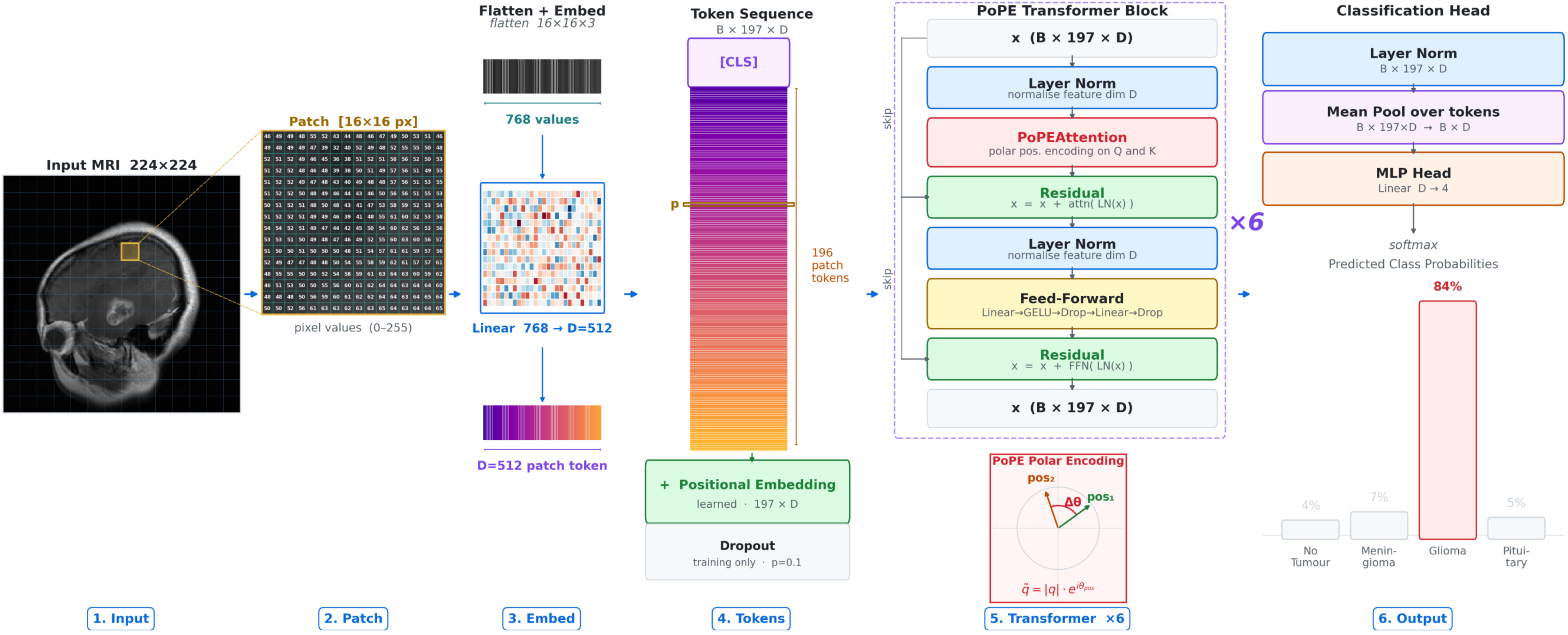
Forward Pass of ViT – Giloma Tumor Example



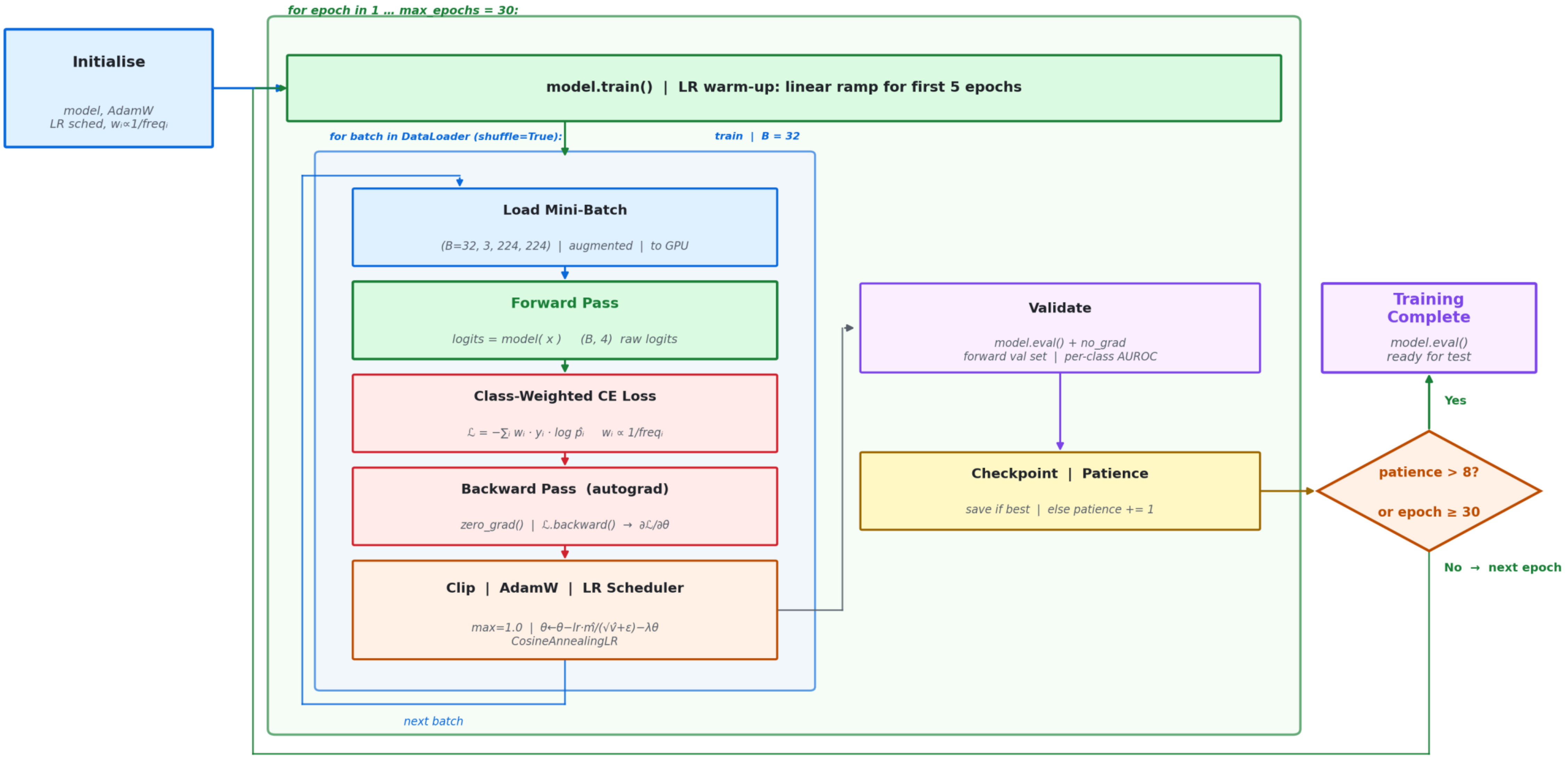
Forward Pass of ViT – Giloma Tumor Example



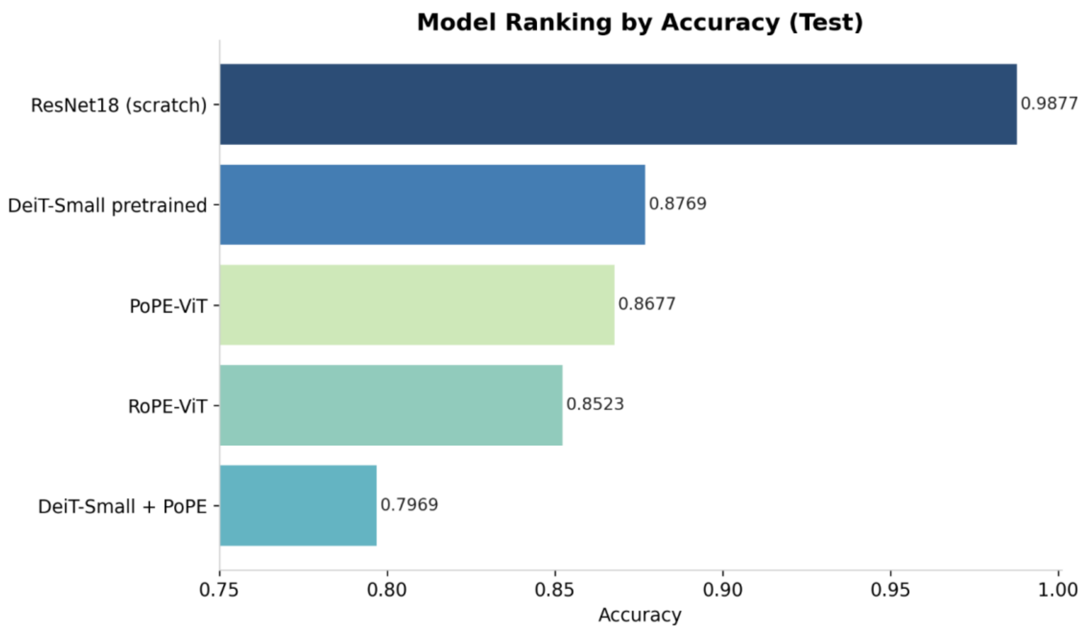
Forward Pass of ViT – Giloma Tumor Example



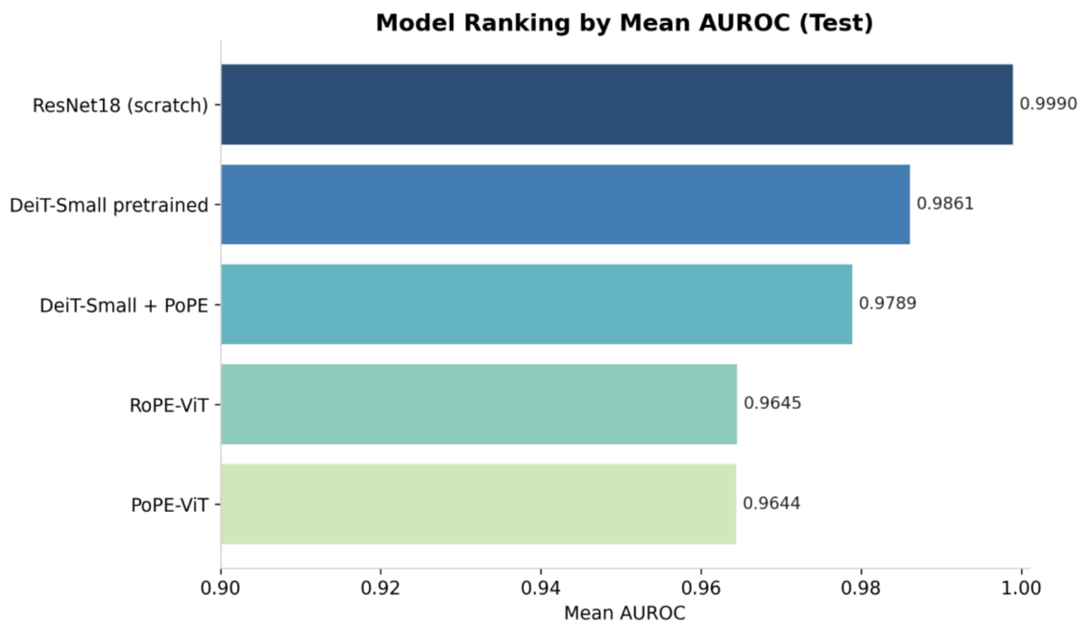
Training Pipeline



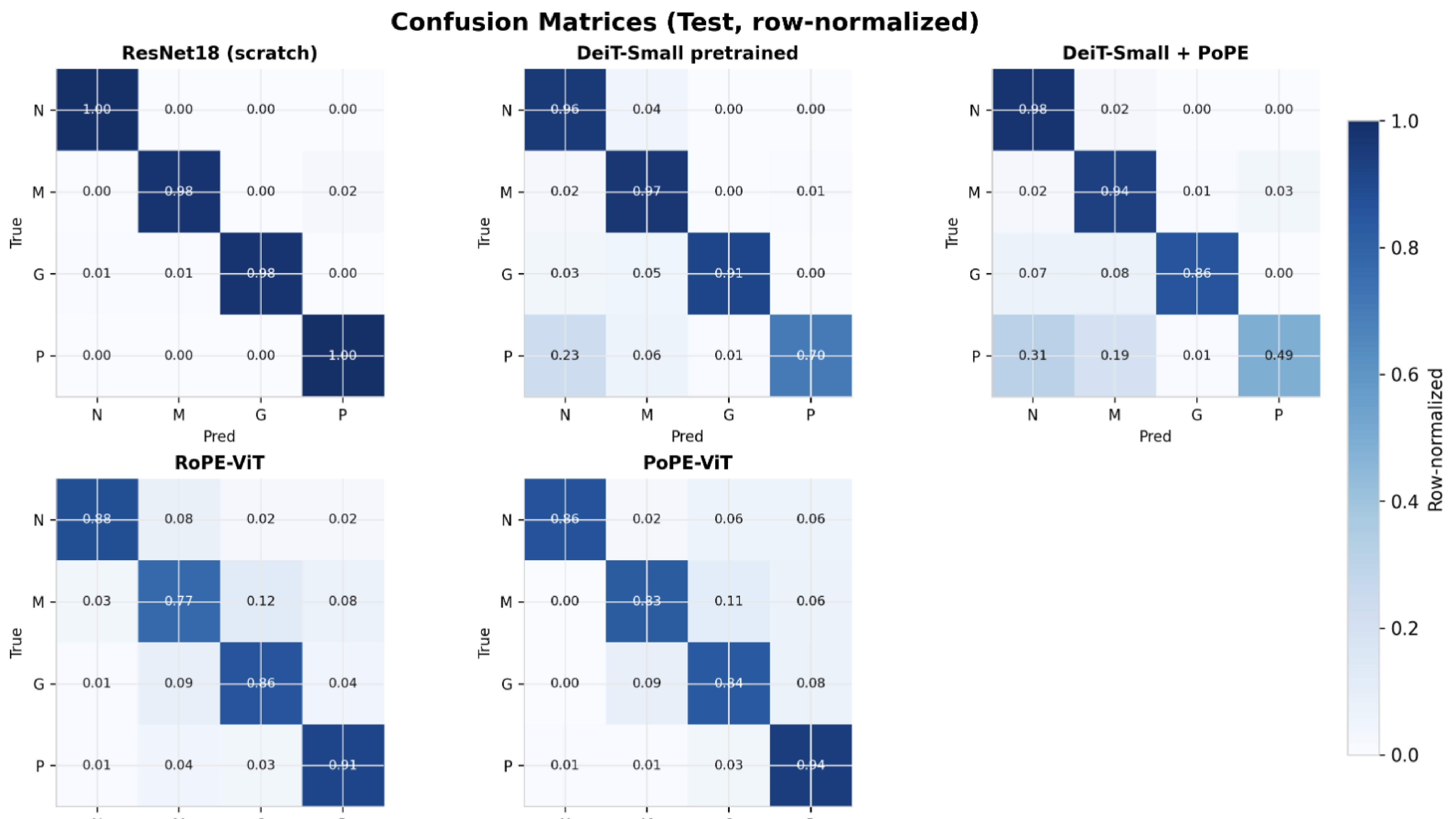
Results



Model Ranking by Accuracy (Test)
ResNet18 (scratch)
DeiT-Small pretrained
PoPE-ViT
RoPE-ViT
DeiT-Small + PoPE
0.9877
0.8769
0.8677
0.8523
0.7969
0.75
0.80
0.85
0.90
0.95
1.00
Accuracy



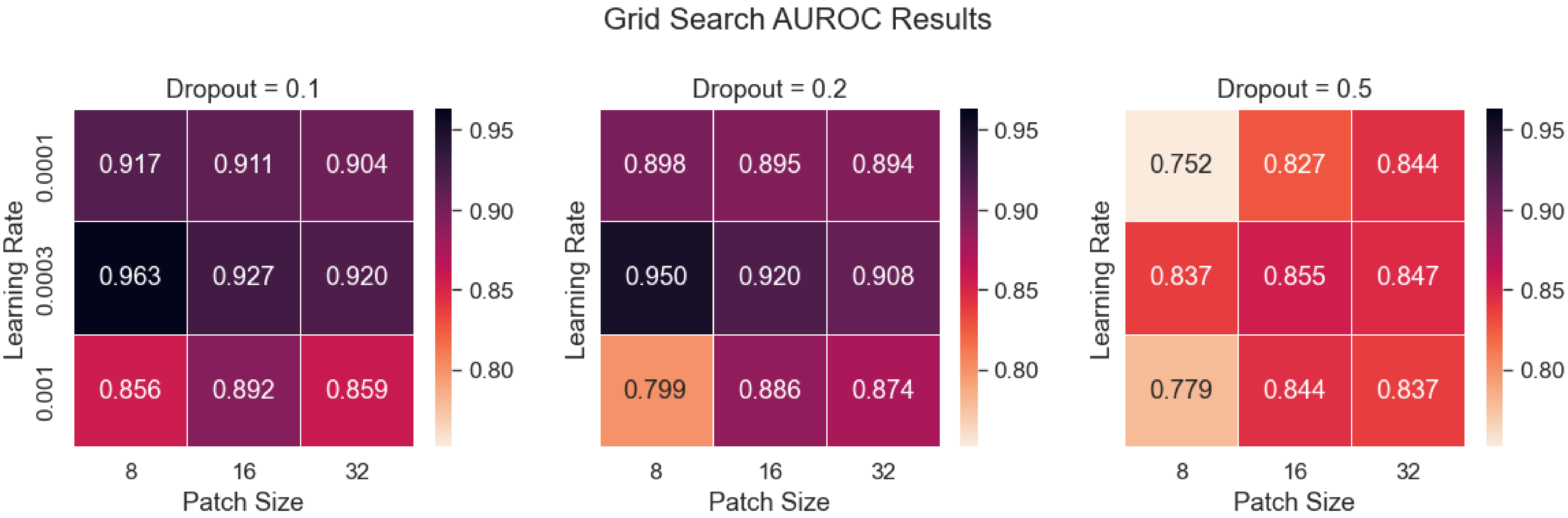
Model Ranking by Mean AUROC (Test)
ResNet18 (scratch)
DeiT-Small pretrained
DeiT-Small + PoPE
RoPE-ViT
PoPE-ViT
0.9990
0.9861
0.9789
0.9645
0.9644
0.90
0.92
0.94
0.96
0.98
1.00
Mean AUROC


Confusion Matrices (Test, row-normalized)
ResNet18 (scratch)
DeiT-Small pretrained
DeiT-Small + PoPE
RoPE-ViT
PoPE-ViT
True
Pred
N
M
G
P
1.00
0.00
0.98
0.02
0.01
0.96
0.04
0.97
0.03
0.05
0.91
0.23
0.06
0.70
0.94
0.07
0.08
0.86
0.31
0.19
0.49
0.88
0.77
0.12
0.09
0.91
0.83
0.11
0.84
Row-normalized
0.0
0.2
0.4
0.6
0.8
1.0

Results: Hyperparameter Tuning

Grid search over patch size, dropout and learning rate:

- Best setting: Patch size: 8, Dropout: 0.1, Learning rate: 3e-4
- Performance decreases with increasing dropout percentage
- Medium learning rate of 3e-4 was ideal
- Patch size has different effects at different learning rates



Limitations

- A key limitation is that our core custom ViT experiments (RoPE-ViT and PoPE-ViT) were trained from scratch rather than initialized from a large-scale pretrained visual transformer backbone, which may have limited their final performance and generalization.
- One limitation of the dataset is that multiple MRI slices originate from the same patient. Since the dataset does not provide patient-level identifiers, the data split was performed at the image level. Consequently, images from the same patient may appear in both training and testing sets, which could lead to optimistic performance estimates.

The Intuition behind PoPE: Mathematical Derivation

Core observation of Gopalakrishnan et al. 2025:

In RoPE, for a query vector q at time step t and a key vector k at time step s, every component q_{tc} and k_{sc} at component index c can be expressed as a complex number in polar form:

$$q_{tc} \Leftrightarrow (\mu_{q_{tc}}, \phi_{q_{tc}})$$

$$k_{sc} \Leftrightarrow (\mu_{k_{sc}}, \phi_{k_{sc}})$$

The attention score between the RoPE-embedded vectors is then computed as the real part of the complex inner product between $RoPE(q_{tc})$ and $RoPE(k_{sc})$:

$$a_{ts}^{RoPE} = Re(\overline{RoPE(q_{tc})} RoPE(k_{sc}))$$

Which simplifies to

$$a_{ts}^{RoPE} = \mu_{q_{tc}} \mu_{k_{sc}} cos((s-t)\theta_c + \phi_{k_{sc}} - \phi_{q_{tc}})$$

The first term $(\mu_{q_{tc}} \mu_{k_{sc}})$ represents the similarity between the content of the two vectors. Ideally we would want the second term to capture only the similarity between positions (i.e. the relative position $(s-t)$), but it also depends on the initial angle $\phi_{q_{tc}}$ and $\phi_{k_{sc}}$, which is a function of the content of the vectors.

In PoPE, the initial initial angles are set to 0, so that the attention score simplifies to:

$$a_{ts}^{PoPE} = \mu_{q_{tc}} \mu_{k_{sc}} cos((s-t)\theta_c)$$

Source: Gopalakrishnan, A., Csordás, R., Schmidhuber, J., & Mozer, M. C. (2025). Decoupling the ‘what’ and ‘where’ with polar coordinate positional embeddings. arXiv (Preprint). https://doi.org/10.48550/ARXIV.2509.10534